



# TARGET-90

JKBOSE CLASS-11<sup>TH</sup>

**SOME BASIC CONCEPTS  
OF CHEMISTRY**

One Shot

**CHEMISTRY**

*Vishal Sir*

Bhaichara  
ON  
TOP



# Topics To Be Covered

1 ... One Shot

2 ...

3 ...

4 ...

# JKBOSE SYLLABUS.

## UNIT-1: SOME BASIC CONCEPTS OF CHEMISTRY

07 Marks

General Introduction: Importance of studying chemistry, Historical approach to particulate nature of matter, Laws of Chemical combination (numerical), Dalton's Atomic Theory, Concept of elements, atoms & molecules. Atomic and molecular masses, Mole concept and molar mass, percentage composition, empirical and molecular formula; chemical reactions, stoichiometry and calculation based on stoichiometry.

CONTENT → CRISP.

# # SCOPE AND IMPORTANCE OF CHEMISTRY

- Health Sector

  - Many life saving drugs. e.g. Cis-Platin (Anticancer), Taxol and Azidothymidine (Treatment of AIDS).
- Warfare

  - So many Explosive e.g. Dynamite, Nitroglycerine, TNT, Atom bomb.
- Food and Preservatives

  - Fertilisers like Urea, weedicides, Pesticides, fungicides etc.
  - So many preservatives.
- Atomic & Nuclear Energy

  - Alternate source of energy i.e. Atomic and Nuclear Energy
- Saving of Environment

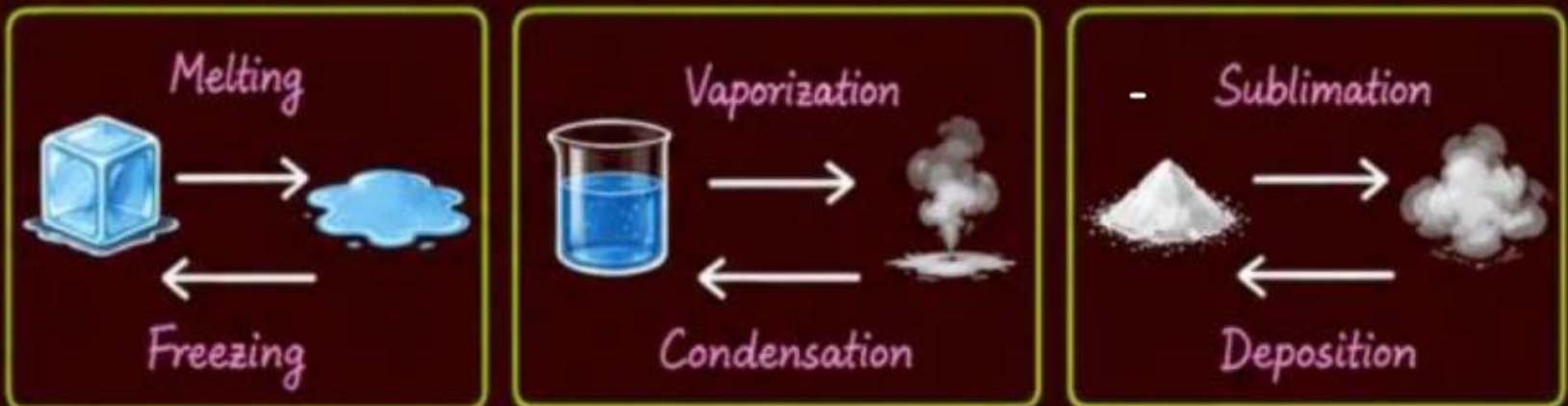
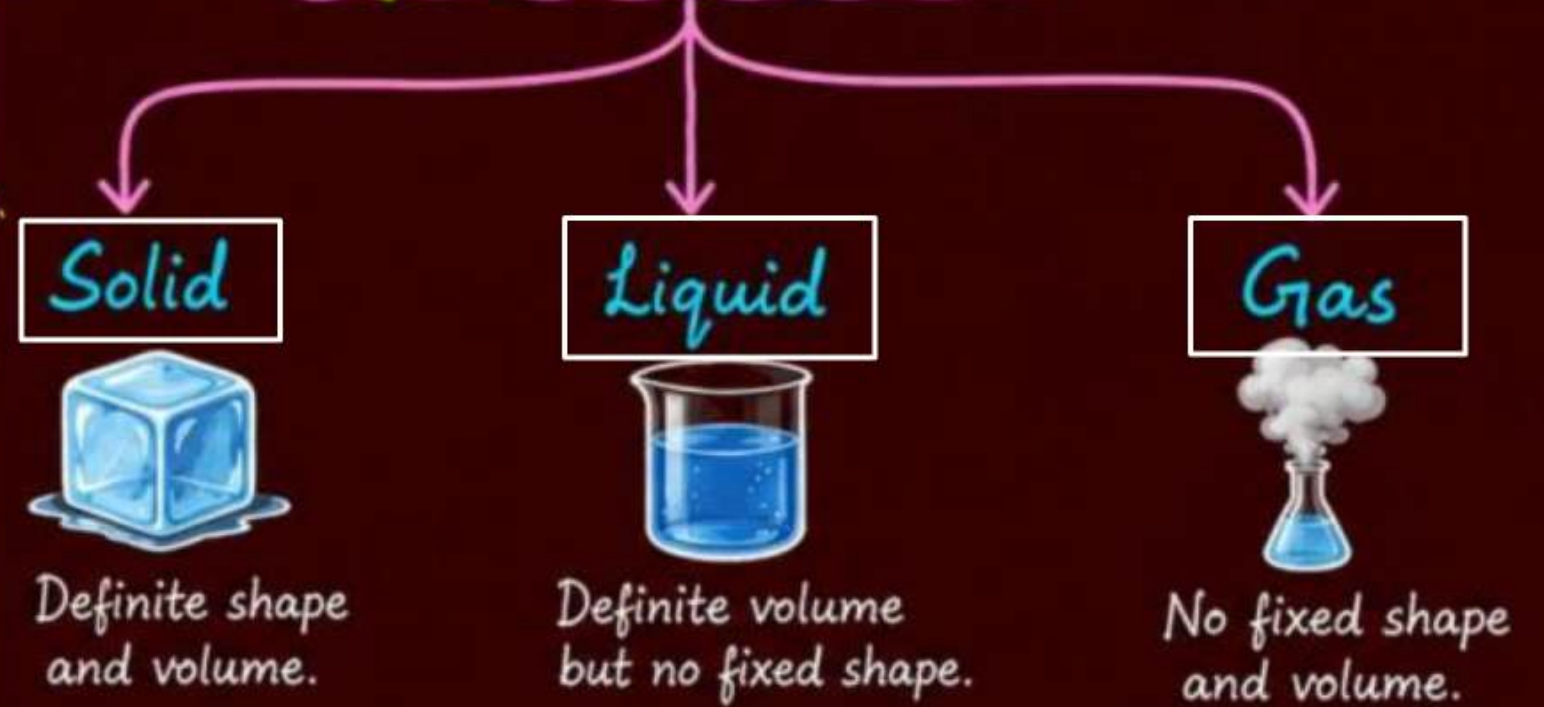
  - Refrigerants like CFC (leads O<sub>3</sub> layer depletion) replaced by HFC.
- Transport

  - Almost all means of transport run on petrol, diesel, CNG etc all are chemicals.
- Nations development

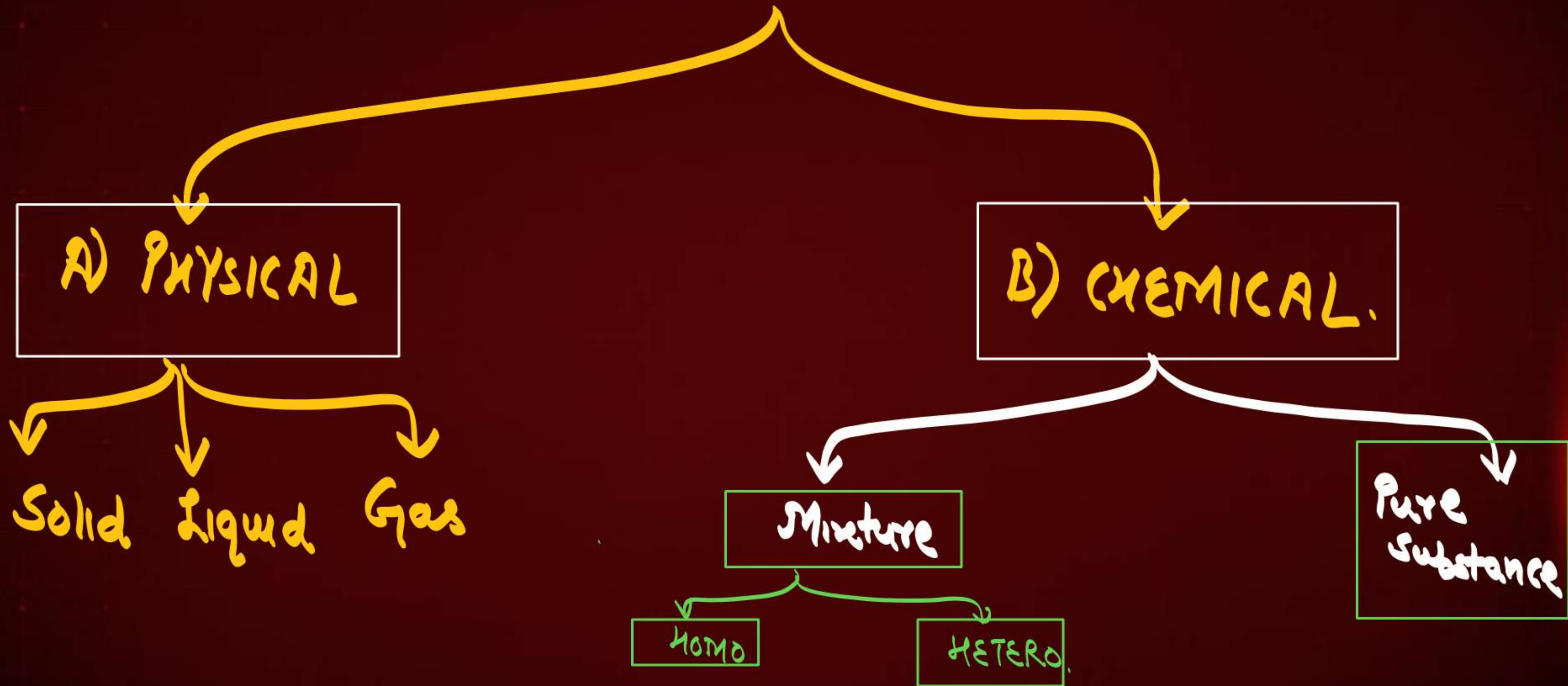
  - Has provides many substances which help in development of Nation (e.g. Polymers, Optical fibres etc).

## Classification of matter.

### Physical Classification



# Matter



H  
W  
F  
A  
S  
T  
N.

Q. Which of the following drug is used in treatment of AIDS?

a) TAXOL

b) CIS-PLATIN

c) AZIDOTHYMIDINE

d) Analgesics.

# # SCOPE AND IMPORTANCE OF CHEMISTRY

## 1) Health Sector



Chemistry helps in the discovery of medicines and vaccines.

## 2) Warfare



Used in the development of advanced weapons, explosives and protective gear.

## 3) Food



Chemistry is used in food processing, packaging, preservation and nutrition.

## 4) Atomic & Nuclear Energy



Chemistry helps in the production and safe use of atomic energy.

## 5) Saving of Environment



Chemistry helps in pollution control, waste management and finding clean energy solutions.

## 6) Transport



Chemistry improves fuel efficiency, battery technology and materials for vehicles.

## 7) Nations development



Chemistry contributes to industries, agriculture, technology and overall economic growth.

## Classification of matter.

### Physical Classification

#### Solid



Definite shape and volume.

#### Liquid



Definite volume but no fixed shape.

#### Gas



No fixed shape and volume.



#### Melting



#### Freezing

#### Vaporization



#### Condensation

#### Sublimation



#### Deposition



# Chemical Classification



## Mixtures

Two or more substances are mixed in any ratio.  
e.g Sugar solution



Can be separated by physical methods.

### Homogeneous



Uniform composition.

sugar solution, Alloys.

i.e. Different components are not visible.

### Heterogeneous



Non-uniform composition.

Components visible.

Fe



## Pure Substances

Type of matter having one kind of particle



### Elements

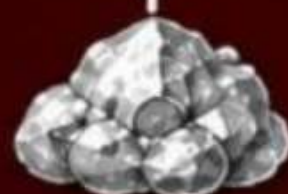
one kind of atom.  
Fe, Cu, Ag etc.



Metal



Non-metal



Metalloids

### Compounds

Two or more different elements present in a fixed ratio.  
e.g.  $H_2O$ ,  $CO_2$ ,  $C_6H_{12}O_6$



Organic

Source living things



Inorganic

Source non-living

Q. Which of the following compound is Organic?

a) NaCl

b) Milk

c)  $C_6H_{12}O_6$

d)  $CaCO_3$

↓  
Mixture

↓  
HETEROGENOUS

~~Ans →  
Milk~~

## Dalton's Atomic Theory

Given by Dalton.

## Limitations

## Dalton's Atomic Theory

2025 2 marks

1. Matter is made of tiny, indivisible, indestructible particles called atoms.
2. Atoms of the same element are identical in shape, size, mass, and properties.
3. Atoms of different elements differ in mass and properties.
4. Atom is the smallest unit in chemical reactions.
5. Atoms combine in fixed ratios to form molecules.
6. Atoms cannot be created or destroyed.
7. Chemical reactions only involve rearrangement of atoms.



H.W

## Limitations of Dalton's Theory

1. Could not explain gaseous laws (law of volumes).
2. Could not explain why atoms of different elements vary in mass, size, or valency.
3. Could not explain why atoms combine to form molecules.
4. Could not explain the forces that hold atoms together in molecules.

⑤ Isotopes ✓

⑥ Isobars

# Laws of Chemical COMBINATION

Important

Father of CHEMISTRY.

## ① Law of Conservation of mass.

• Lavoisier

- In any phy or chem. change mass of reactants is always equal to mass of product.



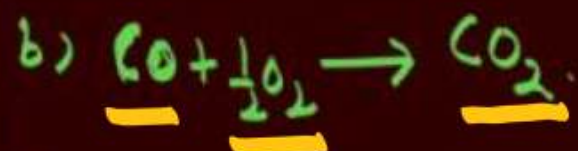
Mass of Reactants  
=  
Mass of Products

• Not followed by Nuclear reaction.

## ② Law of constant composition. (J Proust).

A chemical compound is always formed to be made up of same elements present in a fixed ratio by mass.

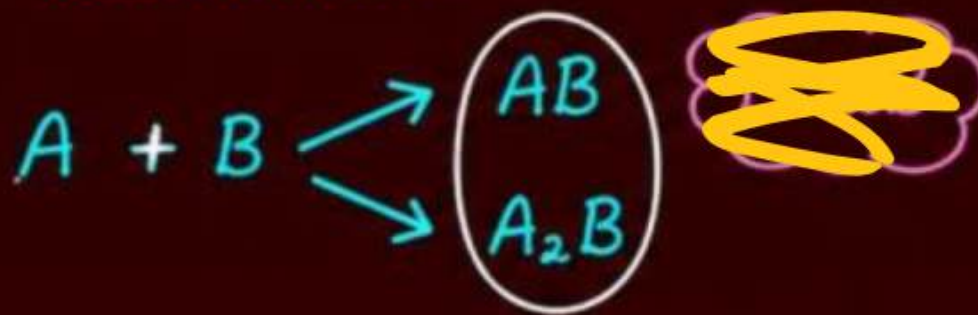
e.g.  $\text{CO}_2$  made from different processes always contain fixed ratio of C & O mass



Both  $\text{CO}_2$  has C & O present in a ratio by mass 12:32 or 3:8

## ③ Law of multiple proportion. (Dalton).

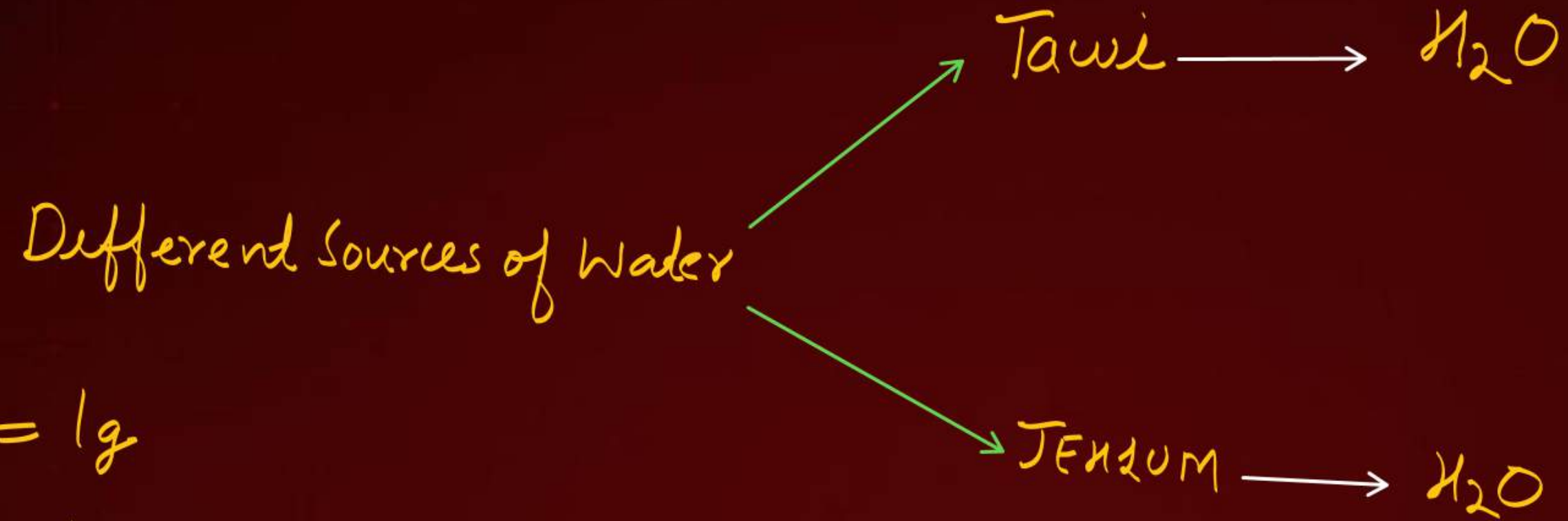
If two elements combine to form more than one compound, the weight of one of the elements that combine with fixed mass of other element bear a simple whole no. ratio.



A = Element 1

B = Element 2





H = 1g  
O = 16g

Ratio by mass  
H : O  
2 : 16 or 1 : 8

e.g Physical change  $\Rightarrow$



e.g CHEMICAL CHANGE  $\Rightarrow$



e.g.

Hydrogen + Oxygen



Here ratio of mass



$$16 : 32$$

$$\boxed{1 : 2}$$

$$1.5 : 2 \times$$

e.g

Phosphorous + Chlorine



Q. HCl prepared from two different experiment & shows fixed ratio of H : Cl by mass

a) Law of Conservation of mass

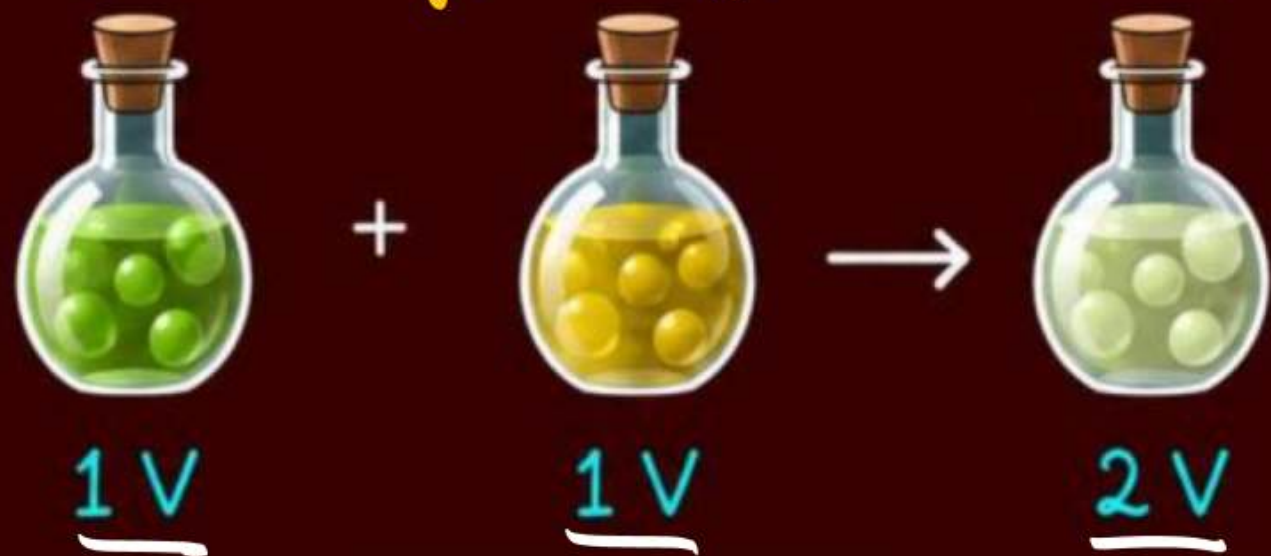
b) Law of constant composition (definite)

c) Gay Lussac.

d) Law of multiple proportion

#### ④ Gay Lussac's law $\Rightarrow$ (Gases).

When gases combine or produced in a chemical reaction, they do so in simple whole no. of ratio by volume.



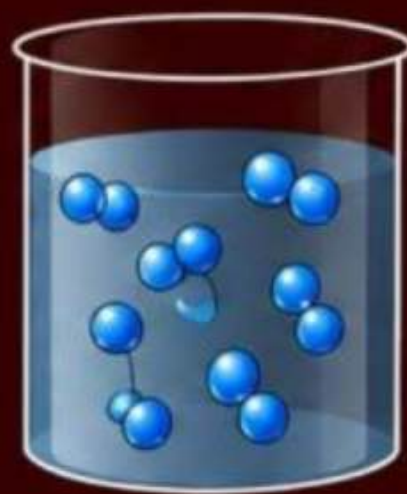
$1 : 1 : 2$  ✓

#### ⑤ Avogadro law $\Rightarrow$

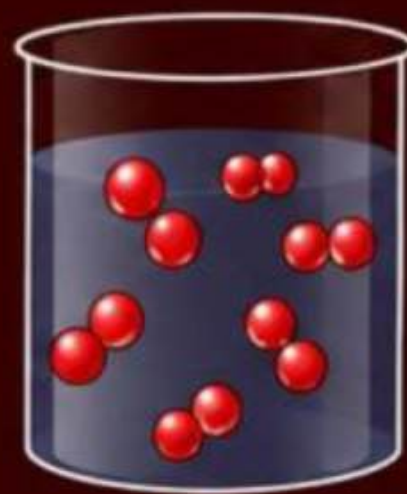


Equal volume of gases contain equal no. of molecules at same temp & pressure.

$$V \propto N$$



1 L  $\text{O}_2$   
at same  
T & P



1 L  $\text{N}_2$   
at same  
T & P

=

Meaning:

Equal volume of  $\text{O}_2$  and  $\text{N}_2$  gases have equal number of molecules.

Q. If  $H_2$ ,  $Cl_2$  &  $O_2$  has same volume & T & P. same, if  $H_2$  &  $Cl_2$  has 50 molecules then  $O_2$  has.

- a) 10      b) 25      c) 50      d) 100.

## Relative Atomic Mass



$$RAM = \frac{\text{Mass of that isotope}}{\frac{1}{12} \text{ of mass of } {}^{12}_6\text{C}}$$

$$\frac{1}{12} \text{ of mass of } {}^{12}_6\text{C} = 1 \text{ amu.}$$

$$1 \text{ amu} = 1 \text{ u}$$

unified mass

$$1 \text{ amu} = 1.66 \times 10^{-24} \text{ g.}$$



${}^{12}_6\text{C}$



$\frac{1}{12} \text{ th}$

$$= 1 \text{ amu (u)}$$



## ★ Important Average atomic mass.

${}^1\text{H}$   ${}^2\text{H}$   ${}^3\text{H}$

For isotopes.

- i) Relative abundance
- ii) Relative Atomic mass

$$AAM = \frac{(RA \times RAM)}{100}$$

$$= (?)$$



${}^1\text{H}$

75%



${}^2\text{H}$

15%



${}^3\text{H}$

10%

## Molecular mass.



$$\text{CH}_4 = 12 + 4 \times 1 = 16 \text{ u}$$



$$\text{H}_2\text{O} = 2 \times 1 + 16 = 18 \text{ u}$$



$$\text{NH}_3 = 14 + 3 \times 1 = 17 \text{ u}$$



$$\begin{aligned} \text{H}_2\text{SO}_4 &= 2 + 32 + 16 \times 4 \\ &= 2 + 32 + 64 \\ &= 34 + 64 \\ &= 98 \text{ u} \end{aligned}$$

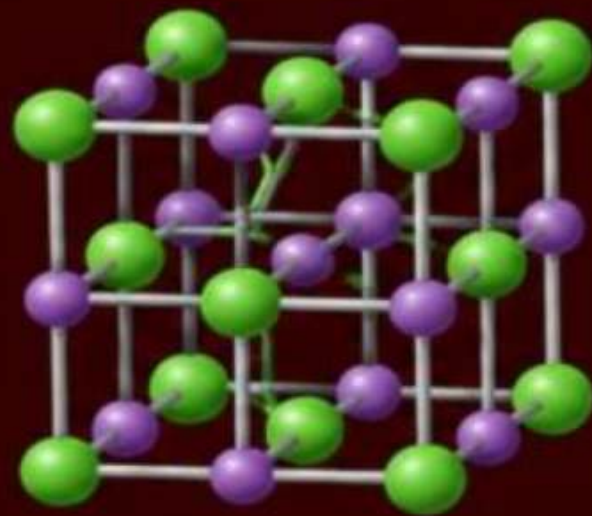
## Formula mass.

For ionic compounds  
or  
solid state.

e.g. NaCl

$$= 23 + 35.5$$

$$= 58.5 \text{ u}$$



$$\text{CO}_2 = 12 + (2 \times 16) = 12 + 32 = 44 \text{ amu.}$$

$$\text{H}_2\text{O} = (2 \times 1) + 16 = 18 \text{ amu.}$$

$$\text{CO} = 12 + 16 = 28 \text{ amu.}$$

$$R.A.M = \frac{\text{Mass of that atom}}{\frac{1}{12}^{\text{th}} \text{ of mass of } ^{12}\text{C}} = \frac{\text{Mass of that atom}}{1 \text{amu} (1.66 \times 10^{-24} \text{g})}$$

## Calculation of average atomic mass

2-3 Marks

- Calculate the atomic mass of naturally occurring argon from the following data:
- Isotope

| Isotope          | Isotopic mass (g mol <sup>-1</sup> ) | Abundance |
|------------------|--------------------------------------|-----------|
| <sup>36</sup> Ar | 35.96755                             | 0.337%    |
| <sup>38</sup> Ar | 37.96272                             | 0.063%    |
| <sup>40</sup> Ar | 39.9624                              | 99.600%   |

$$\begin{aligned}
 \text{A. A. M} &= \frac{(35.96755 \times 0.337) + (37.96272 \times 0.063) + (39.9624 \times 99.600)}{100} \\
 &= 39.9 \text{ g} \cdot \text{mol}^{-1}
 \end{aligned}$$

Calculator  
Allowed hq.

RELATIVE ATOMIC MASS  $\rightarrow$  It is defined as  
 how many times an atom of any element  
 is heavier than  $\frac{1}{12^{\text{th}}}$  of mass of an atom  
 of  $^{12}\text{C}$ .

Element  
(atom)

$\frac{1}{12^{\text{th}}}$  of  $^{12}\text{C}$

5

| Element Name | Atomic Number | Atomic Mass | Element Name | Atomic Number | Atomic Mass |
|--------------|---------------|-------------|--------------|---------------|-------------|
| Hydrogen     | 1             | 1           | Sodium       | 11            | 23          |
| Helium       | 2             | 4           | Magnesium    | 12            | 24          |
| Lithium      | 3             | 7           | Aluminium    | 13            | 27          |
| Beryllium    | 4             | 9           | Silicon      | 14            | 28          |
| Boron        | 5             | 11          | Phosphorus   | 15            | 31          |
| Carbon       | 6             | 12          | Sulphur      | 16            | 32          |
| Nitrogen     | 7             | 14          | Chlorine     | 17            | 35.5        |
| Oxygen       | 8             | 16          | Argon        | 18            | 40          |
| Fluorine     | 9             | 19          | Potassium    | 19            | 39          |
| Neon         | 10            | 20          | Calcium      | 20            | 40          |

$$C = 12 \text{ amu}$$

$$H = 1 \text{ amu}$$

$$O = 16 \text{ amu}$$

$$N = 14 \text{ amu}$$

$$S = 32 \text{ amu}$$

$$Na = 23 \text{ amu}$$

$$Cl = 35.5 \text{ amu.}$$

$$Ca = 40 \text{ amu}$$

$$K = 40 \text{ amu}$$

$$Ar = 39 \text{ amu.}$$

## Gram atomic mass

RAM expressed in gram.



$$\begin{aligned} H &= 1 \text{ amu or } \mu \\ H &= 1 \text{ g} \end{aligned}$$

Molar mass. Mass of one mole of substance.



$$\begin{aligned} \underline{CO_2} &= 12 + 2 \times 16 \\ &= 12 + 32 \\ &= \underline{44 \text{ g mol}^{-1}} \end{aligned}$$

## Mole concept

Amount of substance containing  $6.022 \times 10^{23}$  particles is called mole.

$$1 \text{ mole} = 6.022 \times 10^{23} = \underline{N_A}$$

Avogadro no.

1 mole of  $H_2$

No.

$$6.022 \times 10^{23}$$



Mass

$$2 \text{ g mol}^{-1}$$



Volume

$$22.4 \text{ L}$$



## Important formulas.



$$\text{No. of moles of atom} = \frac{\text{Given AM}}{\text{Molar mass}}$$

$$\text{No. of moles of molecules} = \frac{\text{Given MM}}{\text{Molar mass}}$$

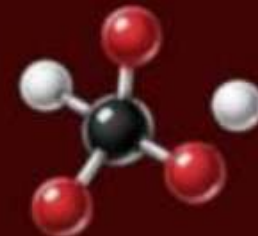
$$\text{No. of moles of Gas} = \frac{\text{Given volume}}{22.4 \text{ L}}$$

$$\star \text{ No. of molecules} = \text{No. of moles} \times N_A$$

$$\star \text{ No. of atoms} = \text{No. of molecules} \times \text{Atoms in molecule (Atomicity)}$$

$$\text{Mass of one atom} = \frac{\text{Atomic Mass}}{N_A}$$

$$\text{" " one molecule} = \frac{\text{Molecular Mass}}{N_A}$$



1 mole of  $H_2 \rightarrow 6.022 \times 10^{23}$  molecule of  $H_2 \rightarrow 2 \text{ g mol}^{-1} = \text{Gram molecule of } H_2.$

1 mole of  $H$ -atom  $\rightarrow 6.022 \times 10^{23}$  atoms  $\rightarrow 1 \text{ g mol}^{-1}$  Gram atom of  $H.$

1 mole of Vishal <sub>Sir</sub>  $\rightarrow 6.022 \times 10^{23}$  Vishal

1 mole of  $CO_2$  molecule  $\rightarrow 6.022 \times 10^{23}$  molecule  $\rightarrow 12 + 22 = 44 \text{ g mol}^{-1} \rightarrow \text{Gram molecule of } CO_2.$

$$6.022 \times 10^{23}.$$

1 mole of  $H_2O(g)$

No.

$6.022 \times 10^{23}$  molecule  
of  $H_2O$

Fix

Molar mass

$18 \text{ g mol}^{-1}$

Volume.

$22.4 \text{ L}$

Fix

1 mole  $\text{NH}_3(\text{g})$   
 $14 + 3 = 17$

No.

$6.022 \times 10^{23}$  molecule

Molar mass

$17 \text{ g mol}^{-1}$

Volume.

$22.4 \text{ L}$

$$1) \text{ No. of moles} = \frac{\text{Given mass}}{\text{Molar mass.}}$$

$$2) \text{ " " moles of gas} = \frac{\text{Given Volume}}{22.4 \text{ L.}}$$

$$3) \text{ No. of molecules} = \text{No. of moles} \times N_A.$$

$$4) \text{ No. of atoms} = \text{No. of molecules} \times \text{Atomcity}$$

$$5) \text{ Mass of one atom} = \frac{\text{Given mass}}{N_A}$$

$$6) \text{ Mass of one molecule} = \frac{\text{Given mass}}{N_A}$$

|                      | Atomcity |
|----------------------|----------|
| $\text{CO}_2$        | = 3      |
| $\text{H}_2\text{O}$ | = 3      |
| $\text{CO}$          | = 2      |
| $\text{H}_2$         | = 2.     |

RAM

Gram atomic mass

Na 23 amu

23 g

Cl 35.5 amu

35.5 g

Ca 40 amu

40 g.

C 12 amu

12 g.

1 mole of  $H_2 = 6.022 \times 10^{23}$  molecules = 2g = Gram molecule of hydrogen =

**QUESTION**

Gram molecule = 1 mol of  $O_2$

The number of gram molecules of oxygen in  $6.02 \times 10^{24}$  CO molecules is:

- 1 10 g molecules
- 2 5 g molecules
- 3 1 g molecule
- 4 0.5 g molecule

## QUESTION

$$\text{No. of atoms} = \text{No. of molecules} \times \text{Atomicity} = \text{No. of moles} \times N_A \times \text{Atomicity}$$



2022

Which of the following contains maximum number of atom?

- 1  $6.023 \times 10^{21}$  molecules of  $\text{CO}_2$

$$\text{No. of atoms} = \frac{6.023 \times 10^{21}}{18 \times 10^{21}} \times 3$$

- 2 22.4 L of  $\text{CO}_2$  at STP

$$\text{No. of atoms} = \frac{\text{Given Volume}}{22.4} \times N_A \times 3 = \frac{22.4}{22.4} \times 3N_A$$

$$\frac{3 \times 6.023 \times 10^{23}}{18 \times 10^{23}}$$

- 3 0.44 g of  $\text{CO}_2$

$$\text{No. of atom} = \frac{\text{Given mass}}{\text{Molar mass}} \times N_A \times 3 = \frac{0.44}{44} \times 3 \times N_A$$

- 4 None of these

## QUESTION

Determine the empirical formula of an oxide on iron which has 69.9% iron and 30.1% oxygen by mass.

[2024]

(Atomic mass of Fe = 56 and O = 16). *Molecular mass = 160 g*

| Element | Given mass | Molar mass | No. of moles | Molar ratio | Simple ratio |  |
|---------|------------|------------|--------------|-------------|--------------|--|
|         |            |            |              |             |              |  |

$$MF = n \times EF$$

$$\therefore n = \frac{MF_{max}}{EF_{max}}$$

=

# PERCENTAGE COMPOSITION

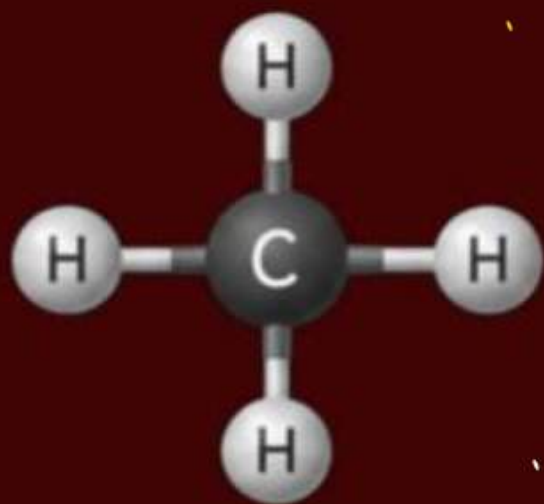
Mass of an element present in 100g of compound.

$$\% = \frac{\text{Mass of that element in a compound}}{\text{Total mass of compound}} \times 100$$



## Molecular Formula.

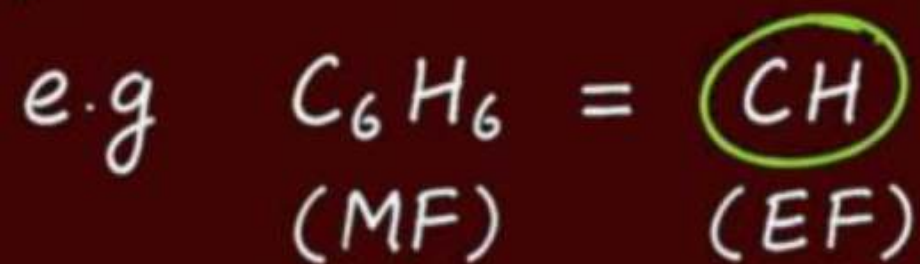
Exact no. of various atoms present in a molecule.



e.g.  $\text{CH}_4$ ,  $\text{C}_8\text{H}_{12}\text{O}_6$   
(Molecular formula of methane)

## Empirical formula.

Ratio of exact no. of different atoms present in a molecules.



## Relationship.

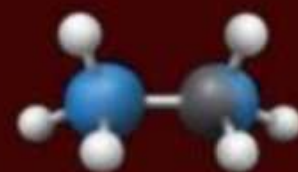
$$\text{MF} = n \times \text{EF}$$

$$n = 1, 2, 3, 4, \dots$$

$$n = \frac{\text{Molecular Formula mass}}{\text{Empirical formula mass}} \checkmark$$

MF = Molecular Formula

EF = Empirical Formula



e.g.  $\text{C}_2\text{H}_6$

= M.W

Find the value of 'n' for  $C_6H_6 =$

$$n = \frac{MF \text{ mass}}{EF \text{ mass}} = \frac{78}{13} = \textcircled{6}$$

$$C_6H_6 (MP) = 78$$

$(12 \times 6) + 6$

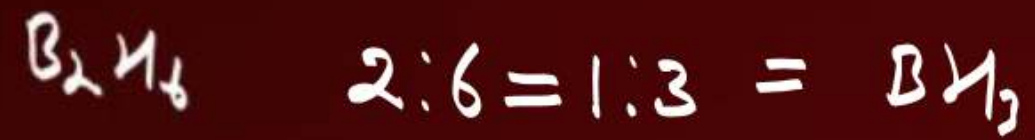
$$CH (EF) = \textcircled{13}$$

$$2 + 16 = 18$$

Q. % composition of H & O in H<sub>2</sub>O

a) % composition of H =  $\frac{2}{18} \times 100 = \checkmark$

b) % composition of O =  $\frac{16}{18} \times 100 =$



STOICHIOMETRY → Relationship between reactants and products in terms of amount in a balanced chemical rxn.

e.g.



|           |            |             |            |             |
|-----------|------------|-------------|------------|-------------|
| Mass      | 16 g       | 64 g        | 44 g       | 36 g        |
| moles     | 1 mole     | 2 moles     | 1 mole     | 2 moles     |
| Molecules | 1 molecule | 2 molecules | 1 molecule | 2 molecules |

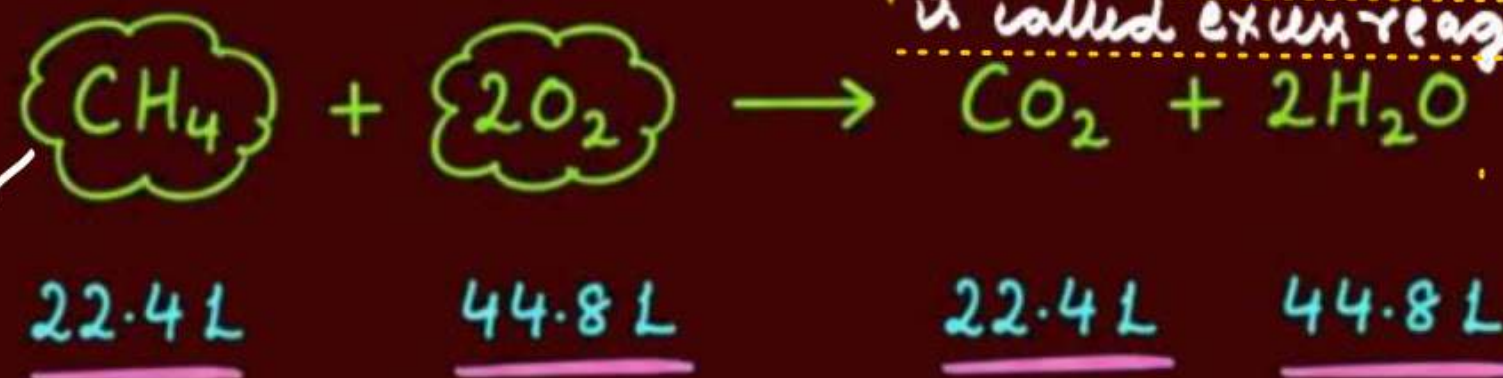
Volume 22.4 L 44.8 L 22.4 L 44.8 L

(at STP)

Limiting Reagent

Reactant  
OR

Reactant which is present in lesser amount, hence limits the formation of products. Other reactant is called excess reagent.



44.7 L → LR

16 g      64 g      44 g      36 g

15 g → LR  
given

Q. Calculate the molecular mass of  $C_6H_{12}O_6$ .

a) 190

b) 180

c) 170

d) 160.

$$\begin{aligned} & (6 \times 12) + (12 \times 1) + (6 \times 16) \\ &= 72 + 12 + 96 \\ &= 180 \text{ g mol}^{-1} \end{aligned}$$

## QUESTION

Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation,



- (i) Calculate the mass of ammonia produced if  $2.00 \times 10^3 \text{g}$  dinitrogen reacts with  $1.00 \times 10^3 \text{g}$  of dihydrogen.
- (ii) Will any of the two reactants remain unreacted? |
- (iii) If yes, which one and what would be its mass?

**Thank**  
*You*